

OU Process

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1 Statement

Suppose $0 < t_1 < t_2 < \dots < t_n < T$, and let $F_i := f(t_i)$ for $i = 1, \dots, n$. If $\Theta = (\sigma^2, \ell, m)$ and $f \mid \Theta = \theta \sim GP(m, k)$, with

$$k(s, t) = \sigma^2 \exp\left(-\frac{|s - t|}{\ell}\right),$$

then the prior can be decomposed as

$$p(F_1, \dots, F_n, \theta) = p(\theta)p(F_1 \mid \theta) \prod_{i=2}^n p(F_i \mid F_{i-1}, \theta).$$

2 Proof

2.1 Variable Setup

Let

$$X_i = F_i - m(t_i), \quad i = 1, \dots, n.$$

Then

$$(X_1, \dots, X_n)^\top \mid \Theta = \theta \sim \mathcal{N}(0, \Sigma),$$

where

$$\Sigma_{ij} = \sigma^2 \exp\left(-\frac{|t_i - t_j|}{\ell}\right).$$

For each $i \geq 2$, define

$$\rho_i = \exp\left(-\frac{t_i - t_{i-1}}{\ell}\right), \quad \epsilon_i = X_i - \rho_i X_{i-1}.$$

2.2 $\epsilon_i \perp (X_1, \dots, X_{i-1}) \mid \Theta = \theta$

Fix $2 \leq i \leq n$ and $1 \leq j \leq i-1$. Write $\text{Cov}_\theta(\cdot, \cdot)$ for covariance conditional on $\Theta = \theta$. Then

$$\begin{aligned}\text{Cov}_\theta(\epsilon_i, X_j) &= \text{Cov}_\theta(X_i - \rho_i X_{i-1}, X_j) \\ &= \text{Cov}_\theta(X_i, X_j) - \rho_i \text{Cov}_\theta(X_{i-1}, X_j).\end{aligned}$$

Since $j \leq i-1 < i$,

$$\begin{aligned}\text{Cov}_\theta(X_i, X_j) &= \sigma^2 \exp\left(-\frac{t_i - t_j}{\ell}\right), \\ \text{Cov}_\theta(X_{i-1}, X_j) &= \sigma^2 \exp\left(-\frac{t_{i-1} - t_j}{\ell}\right).\end{aligned}$$

Moreover,

$$\begin{aligned}\rho_i \text{Cov}_\theta(X_{i-1}, X_j) &= \exp\left(-\frac{t_i - t_{i-1}}{\ell}\right) \sigma^2 \exp\left(-\frac{t_{i-1} - t_j}{\ell}\right) \\ &= \sigma^2 \exp\left(-\frac{t_i - t_j}{\ell}\right) \\ &= \text{Cov}_\theta(X_i, X_j).\end{aligned}$$

Hence,

$$\text{Cov}_\theta(\epsilon_i, X_j) = 0, \quad j = 1, \dots, i-1.$$

Since $(\epsilon_i, X_1, \dots, X_{i-1}) \mid \Theta = \theta$ is jointly Gaussian, zero cross-covariance implies

$$\epsilon_i \perp (X_1, \dots, X_{i-1}) \mid \Theta = \theta.$$

Also, since $(F_1, \dots, F_{i-1})$ is function of $(X_1, \dots, X_{i-1})$,

$$\epsilon_i \perp (F_1, \dots, F_{i-1}) \mid \Theta = \theta.$$

2.3 Markov Property of F_i

Fix $2 \leq i \leq n$. Since

$$X_i = \rho_i X_{i-1} + \epsilon_i,$$

we have

$$F_i - m(t_i) = \rho_i (F_{i-1} - m(t_{i-1})) + \epsilon_i,$$

and hence

$$F_i = m(t_i) + \rho_i (F_{i-1} - m(t_{i-1})) + \epsilon_i.$$

Moreover,

$$\epsilon_i \perp (F_1, \dots, F_{i-1}) \mid \Theta = \theta.$$

Therefore, conditional on $\Theta = \theta$ and F_{i-1} , the only random term in F_i is ϵ_i , which is independent of $(F_1, \dots, F_{i-2})$. Hence,

$$p(F_i \mid F_1, \dots, F_{i-1}, \theta) = p(F_i \mid F_{i-1}, \theta).$$

2.4 Decomposition of the Prior

By the chain rule,

$$p(F_1, \dots, F_n, \theta) = p(\theta) p(F_1 \mid \theta) \prod_{i=2}^n p(F_i \mid F_1, \dots, F_{i-1}, \theta).$$

Using the Markov property,

$$p(F_i \mid F_1, \dots, F_{i-1}, \theta) = p(F_i \mid F_{i-1}, \theta), \quad i = 2, \dots, n.$$

Therefore,

$$p(F_1, \dots, F_n, \theta) = p(\theta) p(F_1 \mid \theta) \prod_{i=2}^n p(F_i \mid F_{i-1}, \theta).$$

3 Induced Distribution

Let

$$X_i := F_i - m(t_i), \quad \rho_i := \exp\left(-\frac{t_i - t_{i-1}}{\ell}\right).$$

Throughout this section, let

$$\mathbb{E}_\theta[\cdot] := \mathbb{E}[\cdot \mid \Theta = \theta],$$

and define Var_θ and Cov_θ analogously.

By the Gaussian process assumption,

$$F_1 \sim_\theta \mathcal{N}(m(t_1), \sigma^2),$$

where $\sim_\theta$ denotes equality in distribution conditional on $\Theta = \theta$.

For $i \geq 2$, define

$$\varepsilon_i := X_i - \rho_i X_{i-1}.$$

Since

$$\mathbb{E}_\theta[X_i] = \mathbb{E}_\theta[X_{i-1}] = 0,$$

we have

$$\mathbb{E}_\theta[\varepsilon_i] = \mathbb{E}_\theta[X_i] - \rho_i \mathbb{E}_\theta[X_{i-1}] = 0.$$

Moreover,

$$\begin{aligned} \text{Var}_\theta(\varepsilon_i) &= \text{Var}_\theta(X_i) + \rho_i^2 \text{Var}_\theta(X_{i-1}) \\ &\quad - 2\rho_i \text{Cov}_\theta(X_i, X_{i-1}). \end{aligned}$$

Under the Matérn kernel with $\nu = \frac{1}{2}$,

$$\text{Var}_\theta(X_i) = \sigma^2$$

and

$$\text{Cov}_\theta(X_i, X_{i-1}) = \sigma^2 \rho_i.$$

Therefore,

$$\text{Var}_\theta(\varepsilon_i) = \sigma^2(1 - \rho_i^2).$$

Since ε_i is a linear combination of jointly Gaussian random variables,

$$\varepsilon_i \sim_\theta \mathcal{N}(0, \sigma^2(1 - \rho_i^2)).$$

Using

$$F_i = m(t_i) + \rho_i(F_{i-1} - m(t_{i-1})) + \varepsilon_i,$$

we obtain

$$F_i \mid F_{i-1} = f_{i-1} \sim_\theta \mathcal{N}(m(t_i) + \rho_i(f_{i-1} - m(t_{i-1})), \sigma^2(1 - \rho_i^2)).$$